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IP Addressing

What is an IP Address?

An IP address is a unique identifier for a device on a network.

Example: 192.168.1.10

IP = Device identity + Location on network

IPv4 = **32 bits → 4 octets**

192.168.1.10

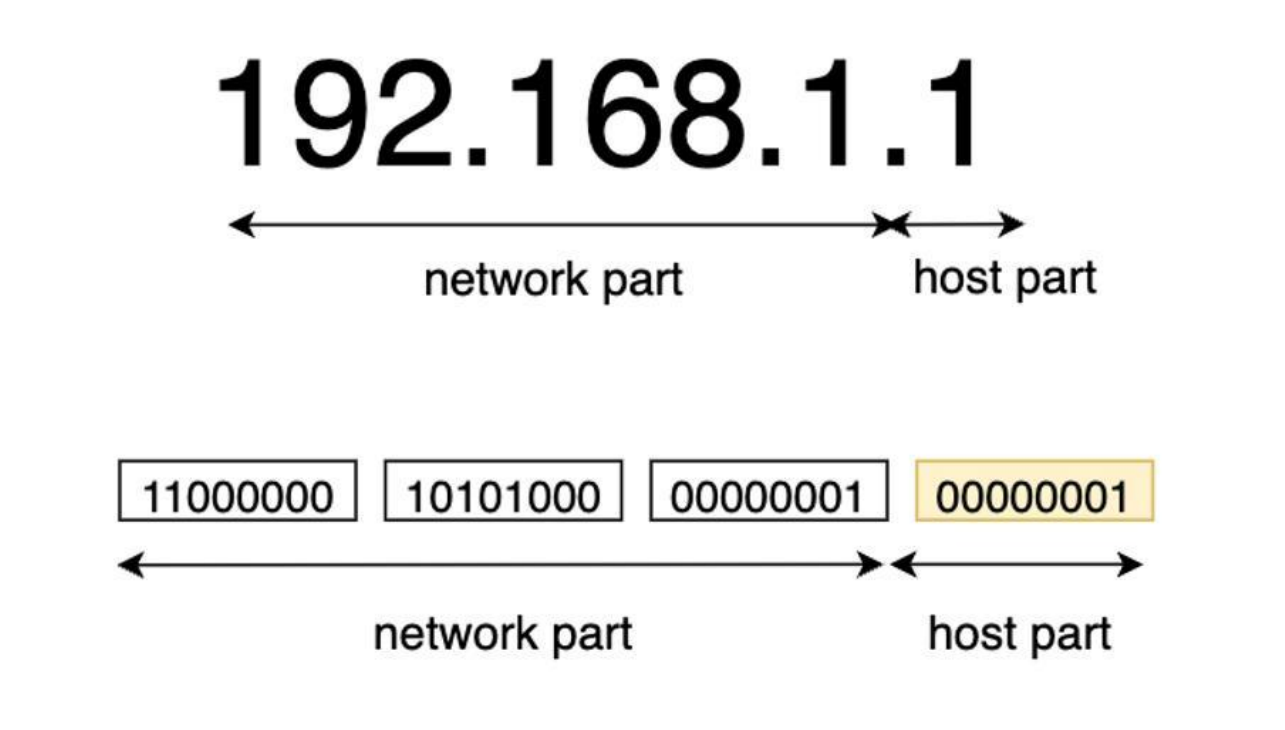
Each part = 8 bits (0–255)

Binary Concept (Important)

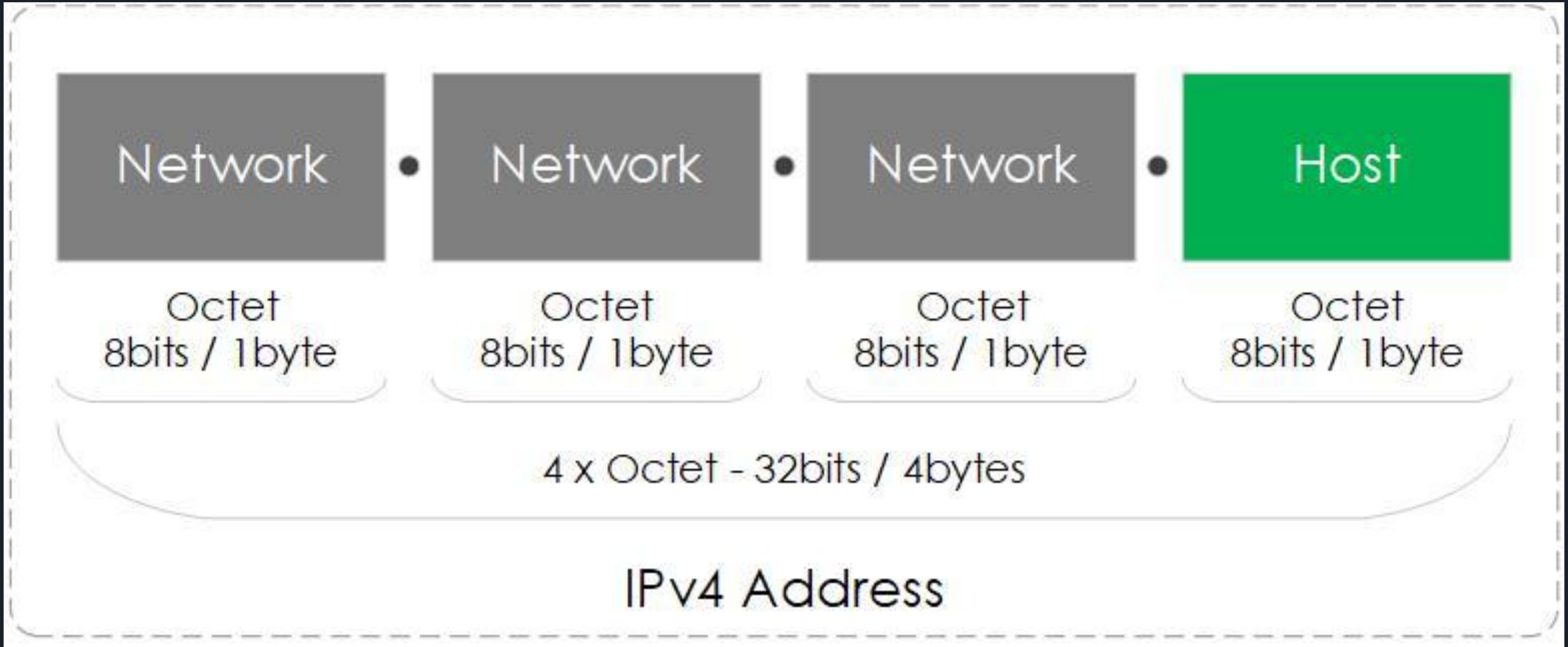
192 → 11000000
168 → 10101000

SOC doesn't use binary daily, but understanding helps in subnetting.

IP Addressing



IP Addressing



Network Part

This identifies **which network** the device belongs to.

Think of it like:

City or Area in an address

Example:

- In `192.168.1.10`, the network part might be: `192.168.1`

This means:

- All devices with `192.168.1.X` are in the **same network**

IP Addressing

Host Part

This identifies the **specific device inside that network**

Think of it like:

House number in a city

Example:

- In `192.168.1.10`, the host part is: `10`

This means:

- Device number 10 in that network

IP Addressing

Subnetting Basics

What is Subnetting?

Subnetting = dividing a network into smaller networks.

Why?

- Better security
- Better performance
- Easier management

CIDR Notation

192.168.1.0/24

/24 means:

- 24 bits = network
- 8 bits = hosts

What is CIDR

CIDR (Classless Inter-Domain Routing). CIDR improves IP address management by allowing more flexible subnetting and efficient allocation of IP addresses, significantly reducing waste.

Organizations use CIDR to allocate IP addresses flexibly and efficiently in their networks.

What is a /24 Subnet?

In IP addressing, the /24 notation refers to a subnet mask with 24 bits dedicated to the network portion of an address. This leaves 8 bits available for the host portion. Subnet masks define how an IP address is divided into network and host segments, which in turn determines the number of IPs available in a given range.

IP Addressing

How Many IP Addresses in a /24 Subnet?

A /24 subnet provides a total of **256 IP addresses**, calculated as follows:

Formula:

Total IPs = 2^(32 – subnet prefix)

Total IPs = 2^(32 – 24) = 2^8 = **256**

Usable IPs:

Out of these 256 IPs, two are reserved:

- **Network Address**: The first IP (e.g., 192.168.0.0), which identifies the subnet itself.
- **Broadcast Address**: The last IP (e.g., 192.168.0.255), used to communicate with all hosts in the subnet.

Thus, a /24 subnet has **254 usable IP addresses** for devices.

IP Addressing

Subnetting Basics

Common Subnets

CIDR	Subnet Mask	Hosts
/24	255.255.255.0	254
/16	255.255.0.0	65,534
/8	255.0.0.0	16M

IP Addressing

Versions of IP Addresses

IPv4 (Internet Protocol Version 4)

- Address size: 32-bit
- Format: Decimal (e.g., 192.168.1.1)
- Total addresses: 4,294,967,296
- IPSec: Optional
- Widely used but facing address exhaustion

IPv6 (Internet Protocol Version 6)

- Address size: 128-bit
- Format: Hexadecimal (e.g., 2001:db8::1)
- Total addresses: 3.4×10^{38}
- IPSec: Mandatory
- Designed to solve IPv4 address shortage

IP Addressing

Classes of IP Addresses

Class A

- Range: 1.0.0.0 – 126.255.255.255
- Network bits: 8
- Host bits: 24
- Use: Large networks
- Example: Government organizations
- 127 is reserved for loopback

Class B

- Range: 128.0.0.0 – 191.255.255.255
- Network bits: 16
- Host bits: 16
- Use: Medium-sized networks
- Example: Universities

Classes of IP Addresses

Class C

- Range: 192.0.0.0 – 223.255.255.255
- Network bits: 24
- Host bits: 8
- Use: Small networks
- Example: Home and small businesses

Class D

- Range: 224.0.0.0 – 239.255.255.255
- Use: Multicast communication
- Example: Video streaming

IP Addressing

Classes of IP Addresses

Class E

- Range: 240.0.0.0 – 255.255.255.255
- Use: Experimental and research
- Not used for public networking

Types of IP Addresses

1. Public
2. Private
3. Static
4. Dynamic

IP Addressing

SOC Perspective

IP Type	Meaning
Private	Internal traffic
Public	External connection

Example

Source: 192.168.1.10
Destination: 45.77.123.66

Internal → External communication

MAC Address

What is MAC Address

A **MAC address** is a **hardware-level unique identifier**.

Example: 00:1A:2B:3C:4D:5E

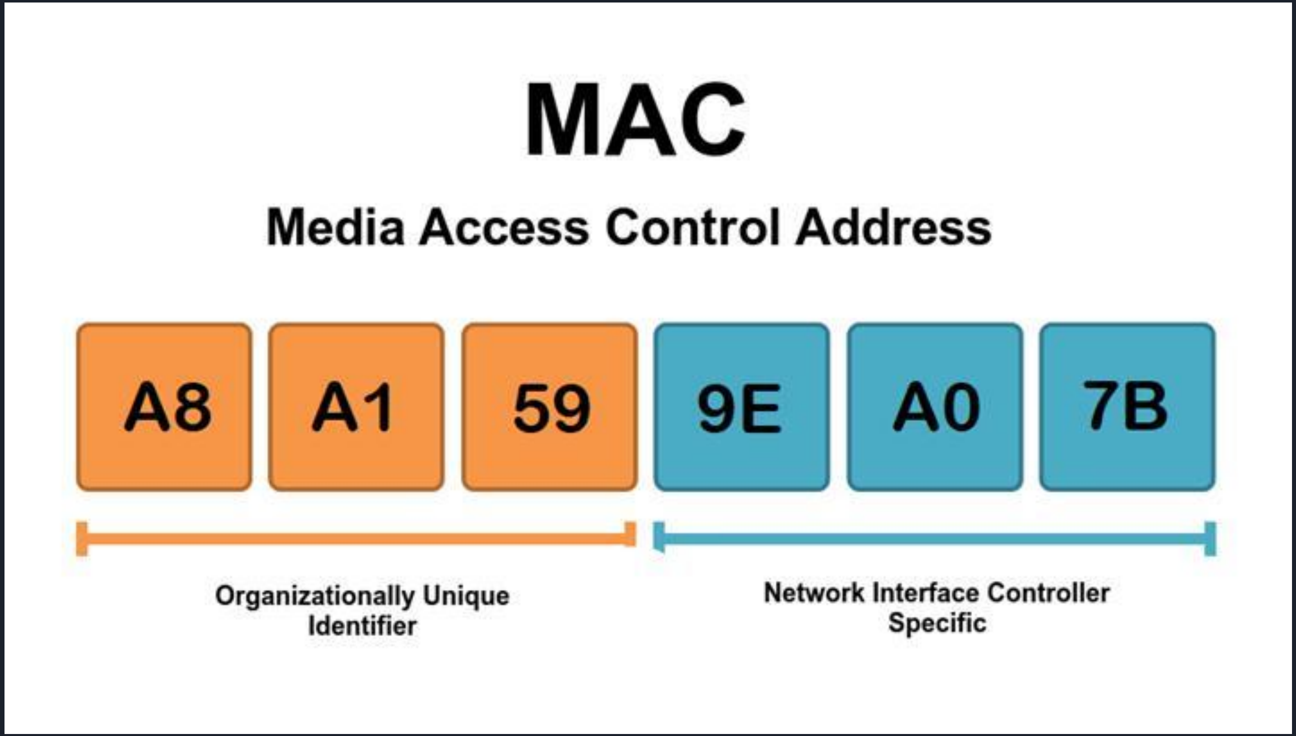
Structure

- 48-bit address
- Written in hex

Parts

- First half → Manufacturer (OUI)
- Second half → Device ID

MAC Address



MAC
Media Access Control Address
A8
A1
59
9E
A0
7B
Organizationally Unique Identifier
Network Interface Controller Specific

MAC Address

ARP - Address Resolution Protocol

ARP maps:

IP → MAC

Example

Who has 192.168.1.1?
Tell me your MAC address

MAC Address

MAC ADDRESS VERSUS IP ADDRESS

MAC ADDRESS	IP ADDRESS
A unique identifier assigned to a Network Interface Controller (NIC) of the computing device	A numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication
Stands for Media Access Control Address	Stands for Internet Protocol Address
Also called physical, hardware or ethernet address	Also called logical address, network or internet address
Helps to uniquely identify the device	Helps to identify the connection of a device on the internet
Assigned by the device manufacturer	Assigned by an administrator or an ISP
Cannot be changed	Can be changed
48 bits (6 bytes) long	IPv4 - 32 bits (4 bytes) IPv6 - 128bits (16bytes)
Works in the Data Link Layer	Works in the Network Layer

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MAC Address

Real SOC Scenario

Multiple devices showing same MAC address

Likely:

ARP Spoofing / MITM